

Joint Probability Distribution

9.5. Joint Distributions. The distribution of two or more random variables which are observed simultaneously when an experiment is performed, is called their *joint distribution*. It is customary to call the distribution of a single r.v. as *univariate*. Likewise, a distribution involving two, three or many r.v.'s simultaneously is referred to as *bivariate*, *trivariate* or *multivariate*.

9.5.1. Bivariate Distribution Function. Let X and Y be two r.v.'s defined on the same sample space S . Then the function $F(x, y)$ defined by $F(x, y) = P(X \leq x \text{ and } Y \leq y)$, where $F(x, y)$ gives the probability that X will take on a value *less than or equal to* x and, at the same time, Y will take on a value *less than or equal to* y , is called a *bivariate or joint distribution function* of X and Y .

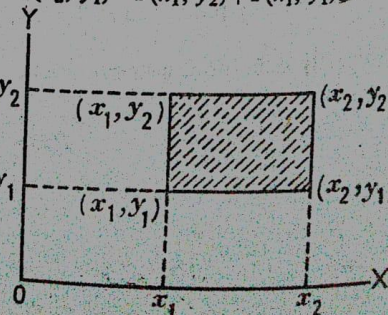
A bivariate d.f. $F(x, y)$ possesses the following properties :

- (i) $F(x, -\infty) = F(-\infty, y) = 0$, $F(+\infty, +\infty) = 1$
- (ii) $F(x, y)$ is a non-decreasing function of x and y , and is continuous on the right.
- (iii) If $x_1 < x_2$ and $y_1 < y_2$, then

$$\begin{aligned} P(x_1 \leq X < x_2 ; y_1 \leq Y < y_2) &= P(X < x_2, Y < y_2) - P(X < x_2, Y < y_1) \\ &\quad - P(X < x_1, Y < y_2) + P(X < x_1, Y < y_1) \\ &= F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) > 0 \end{aligned}$$

The probability that a random point (X, Y) falls in the interval $(x_1 \leq X < x_2 ; y_1 \leq Y < y_2)$ is shown graphically.

A bivariate distribution may be *discrete* when the possible values of (X, Y) are finite or countably infinite. It is *continuous*



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if (X, Y) can assume all values in some non-countable set of the plane. A bivariate distribution is said *mixed* when one r.v. is discrete and the other is continuous.

9.5.2. Bivariate Probability Function. Let X and Y be two discrete r.v.'s defined on the same sample space S , X taking the values $x_1, x_2, \dots, x_m$ and Y taking the values $y_1, y_2, \dots, y_n$. Then the probability that X takes on the value x_i and, at the same time, Y takes on the value y_j , denoted by $f(x_i, y_j)$ or p_{ij} , is defined to be the *joint probability function* or simply the *joint distribution* of X and Y . Thus the *joint probability function*, also called the *bivariate probability function* $f(x, y)$ is a function whose value at the point (x_i, y_j) is given by

$$f(x_i, y_j) = P(X = x_i \text{ and } Y = y_j), \quad \begin{matrix} i=1, 2, \dots, m. \\ j=1, 2, \dots, n. \end{matrix}$$

The joint or bivariate probability distribution consisting of all pairs of values (x_i, y_j) and their associated probabilities $f(x_i, y_j)$, i.e. the set of triples $[x_i, y_j, f(x_i, y_j)]$ can either be shown in a two-way table as below :

Joint Probability Distribution of X and Y

$X \backslash Y$							$P(X=x_i)$
	y_1	y_2	...	y_j	...	y_n	
x_1	$f(x_1, y_1)$	$f(x_1, y_2)$	...	$f(x_1, y_j)$	...	$f(x_1, y_n)$	$g(x_1)$
x_2	$f(x_2, y_1)$	$f(x_2, y_2)$	...	$f(x_2, y_j)$	...	$f(x_2, y_n)$	$g(x_2)$
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
x_i	$f(x_i, y_1)$	$f(x_i, y_2)$	...	$f(x_i, y_j)$	...	$f(x_i, y_n)$	$g(x_i)$
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
x_m	$f(x_m, y_1)$	$f(x_m, y_2)$	...	$f(x_m, y_j)$	...	$f(x_m, y_n)$	$g(x_m)$
$P(Y=y_j)$	$h(y_1)$	$h(y_2)$	...	$h(y_j)$	...	$h(y_n)$	1

or be expressed by means of a formula for $f(x, y)$. The probabilities $f(x, y)$ can be obtained by substituting appropriate values of x and y in the table or formula.

A joint probability function has the following properties :

- $f(x_i, y_j) > 0$, for all (x_i, y_j) , i.e. for $i=1, 2, \dots, m$; $j=1, 2, \dots, n$.
- $\sum_i \sum_j f(x_i, y_j) = 1$

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9.5.3. Marginal Probability Functions. From the joint probability function for (X, Y) , we can obtain the individual probability function of X and Y . Such individual probability functions are called *marginal probability functions*.

Let $f(x, y)$ be the joint probability function of two discrete r.v.'s X and Y . Then the *marginal probability function of X* is defined as

$$\begin{aligned} g(x_i) &= \sum_{j=1}^n f(x_i, y_j) \\ &= f(x_i, y_1) + f(x_i, y_2) + \dots + f(x_i, y_n) \text{ as } x_i \text{ must occur} \\ &\quad \text{either with } y_1 \text{ or } y_2 \text{ or } \dots \text{ or } y_n. \\ &= P(X=x_i); \end{aligned}$$

that is, the individual probability function of X is found by adding over the rows of the two-way table.

Similarly, the *marginal probability function for Y* is obtained by adding over the column as

$$\begin{aligned} h(y_j) &= \sum_{i=1}^m f(x_i, y_j) \\ &= P(Y=y_j) \end{aligned}$$

The values of the *marginal probabilities* are often written in the margins of the joint table as they are the row and column totals in the table. The probabilities in each marginal probability function add to 1.

9.5.4. Conditional Probability Functions. Let X and Y be two discrete r.v.'s with joint probability function $f(x, y)$. Then the *conditional probability function for X given $Y=y$* , denoted as $f(x | y)$, is defined by

$$\begin{aligned} f(x_i | y_j) &= P(X=x_i | Y=y_j) \\ &= \frac{P(X=x_i \text{ and } Y=y_j)}{P(Y=y_j)} \\ &= \frac{f(x_i, y_j)}{h(y_j)}, \quad \text{for } i=1, 2, \dots; j=1, 2, \dots \end{aligned}$$

where $h(y)$ is the marginal probability and $h(y) > 0$. It gives the probability that X takes on the value x_i given that Y has taken on the value y_j . The conditional probability $f(x_i | y_j)$ is non-negative and (for a given fixed y_j) adds to 1 on i and hence is a probability function.

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Similarly, the *conditional* probability function for Y given $X=x$ is

$$\begin{aligned} f(y_j | x_i) &= P(Y=y_j | X=x_i) \\ &= \frac{P(Y=y_j \text{ and } X=x_i)}{P(X=x_i)} \\ &= \frac{f(x_i, y_j)}{g(x_i)}, \quad \text{where } g(x) > 0. \end{aligned}$$

9.5.5. Independence. Two discrete r.v.'s X and Y are said to be *statistically independent*, if and only if, for all possible pairs of values (x_i, y_j) , the joint probability function $f(x, y)$ can be expressed as the product of the two marginal probability functions. That is, X and Y are independent, if

$$\begin{aligned} f(x, y) &= P(X=x_i \text{ and } Y=y_j) \\ &= P(X=x_i) \cdot P(Y=y_j) \quad \text{for all } i \text{ and } j. \\ &= g(x) h(y). \end{aligned}$$

It should be noted that the joint p.f. of X and Y when they are independent can be obtained by multiplying together their marginal probability functions.

Example 9.6. An urn contains 3 black, 2 red and 3 green balls and 2 balls are selected at random from it. If X is the number of black balls and Y is the number of red balls selected, then find

- (i) the joint probability function $f(x, y)$;
- (ii) $P(X+Y \leq 1)$;
- (iii) the marginal p.d. $g(x)$ and $h(y)$;
- (iv) the conditional p.d. $f(x | 1)$, (v) $P(X=0 | Y=1)$, and
- (vi) Are X and Y independent ?

- (i) The sample space S for this experiment contains $\binom{8}{2} = 28$

sample points. The possible values of X are 0, 1 and 2, and those for Y are 0, 1 and 2. The values that (X, Y) can take on are (0, 0), (0, 1), (1, 0), (1, 1), (0, 2) and (2, 0). We desire to find $f(x, y)$ for each value (x, y) .

Now $f(0, 0) = P(X=0 \text{ and } Y=0)$, where the event $(X=0 \text{ and } Y=0)$ represents that neither black nor red ball is selected, implying that the 2 selected are green balls. This event

therefore contains $\binom{3}{0} \binom{2}{0} \binom{3}{2} = 3$ sample points, and

$$\begin{aligned} f(0, 0) &= P(X=0 \text{ and } Y=0) \\ &= \frac{3}{28}. \end{aligned}$$

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3.37 Determine the value of K so that the following function represent joint probability distribution of random variable X and Y

$$f(x, y) = Kxy \quad x = 1, 2, 3$$

$$y = 1, 2, 3$$

$$D.V = X.$$

$$D.V = Y.$$

$x \backslash y$	1	2	3	$g(x)$
1	$(11) \frac{1}{36}$	$(12) \frac{2}{36}$	$(13) \frac{3}{36}$	$\frac{6}{36}$
2	$(21) \frac{2}{36}$	$(22) \frac{4}{36}$	$(23) \frac{6}{36}$	$\frac{12}{36}$
3	$(31) \frac{3}{36}$	$(32) \frac{6}{36}$	$(33) \frac{9}{36}$	$\frac{18}{36}$
$h(y)$	$\frac{6}{36}$	$\frac{12}{36}$	$\frac{18}{36}$	1

$$w = K \cdot T$$

$$\sum_x \sum_y f(x, y) = 1$$

Now

$$f(x, y) = Kxy$$

$$f(x=1, y=1) = K \quad \text{--- (1)}$$

$$f(x=1, y=2) = 2K \quad \text{--- (2)}$$

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$$f(x=1, y=3) = 3k \text{ --- (iii)}$$

$$f(x=2, y=1) = 2k \text{ --- (iv)}$$

$$f(x=2, y=2) = 4k \text{ --- (v)}$$

$$f(x=2, y=3) = 6k \text{ --- (vi)}$$

$$f(x=3, y=1) = 3k \text{ --- (vii)}$$

$$f(x=3, y=2) = 6k \text{ --- (viii)}$$

$$f(x=3, y=3) = 9k \text{ --- (ix)}$$

adding $\textcircled{i}$ to $\textcircled{ix}$ we get

$$= 36k.$$

Now

$$\sum \sum f(x, y) = 1$$

$$36k = 1$$

$$\boxed{k = 1/36}.$$

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Marginal probability density function of X and Y

X	$g(x)$	Y	$h(y)$
1	$6/36$	1	$6/36$
2	$12/36$	2	$12/36$
3	$18/36$	3	$18/36$
$\sum g(x) = 1$		$\sum h(y) = 1$	

i) $P(X+Y=4) = f(1,3) + f(2,2) + f(3,1)$
 $= \frac{3}{36} + \frac{4}{36} + \frac{3}{36}$
 $P(X+Y=4) = \frac{10}{36}$

ii) $P(X > Y) = f(2,1) + f(3,1) + f(3,2)$
 $= \frac{2}{36} + \frac{3}{36} + \frac{6}{36}$
 $P(X > Y) = \frac{11}{36}$

iii) $P(X > 2, Y \leq 1) = f(3,1)$
 $P(X > 2, Y \leq 1) = \frac{3}{36}$

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Example. The joint Pdf of two discrete r.v.
 X and Y is given by.

$$f(x, y) = \frac{xy^2}{30} \quad x = 1, 2, 3$$

$$y = 1, 2$$

$x \backslash y$	1	2	$g(x)$
1	$(1, 1)$ $1/30$	$(1, 2)$ $4/30$	$5/30$
2	$(2, 1)$ $2/30$	$(2, 2)$ $8/30$	$10/30$
3	$(3, 1)$ $3/30$	$(3, 2)$ $12/30$	$15/30$
$h(y)$	$6/30$	$24/30$	1

$$f(x, y) = \frac{xy^2}{30}$$

$$f(x=1, y=1) = 1/30$$

$$f(x=1, y=2) = 4/30$$

$$f(x=2, y=1) = 2/30$$

$$f(x=2, y=2) = 8/30$$

$$f(x=3, y=1) = 3/30$$

$$f(x=3, y=2) = 12/30$$

a) marginal Pdf

x	$g(x)$	y	$h(y)$
1	$5/30$	1	$6/30$
2	$10/30$	2	$24/30$
3	$15/30$		$\sum h(y) = 1$

$\sum g(x) = 1$

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b) Are X, Y are independent?

$$f(x, y) = g(x) \cdot h(y)$$
$$f(x=1, y=1) = g(x=1) \cdot h(y=1)$$
$$1/30 = 5/30 \times 4/30$$
$$1/30 \neq 5/30 \times 4/30$$

Hence X and Y are independent.

Example: Two refills for a ballpoint pen are selected at random from a box that contains 3 blue, 2 red, and 3 green refills. If X is r.v for number of Blue and Y is r.v for number of red refills selected find

a) joint probability function $f(x, y)$ and table also

b) $P[(X, Y) \in A]$ where
 $A = \{(x, y) / x + y \leq 1\}$

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Blue balls = 3
 red balls = 2
 green balls = 3

$n.v = x = \text{Blue} \quad x = 0, 1, 2,$
 $n.v = y = \text{red} \quad y = 0, 1, 2.$

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}} \rightarrow 2$$

$$f(x=0, y=0) = \frac{\binom{3}{0} \binom{2}{0} \binom{3}{2-0-0}}{\binom{8}{2}} = \frac{3}{28}$$

$$f(x=0, y=1) = \frac{\binom{3}{0} \binom{2}{1} \binom{3}{2-0-1}}{\binom{8}{2}} = \frac{6}{28}$$

$$f(x=0, y=2) = \frac{\binom{3}{0} \binom{2}{2} \binom{3}{2-0-2}}{\binom{8}{2}} = \frac{1}{28}$$

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$0 \leq x + y \leq 2$

$x \backslash y$	0	1	2	$g(y)$
0	$(0,0)$ $3/28$	$(0,1)$ $6/28$	$(0,2)$ $1/28$	$10/28$
1	$(1,0)$ $9/28$	$(1,1)$ $6/28$	$(1,2)$ $1/28$	$15/28$
2	$(2,0)$ $3/28$	$(2,1)$ $1/28$	$(2,2)$ $1/28$	$3/28$
$h(x)$	$15/28$	$12/28$	$1/28$	1

x	$g(x)$
0	$10/28$
1	$15/28$
2	$3/28$
$\sum g(x) = 1$	

y	$h(y)$
0	$15/28$
1	$12/28$
2	$1/28$
$\sum h(y) = 1$	

$$P[(X,Y) \in A] = P(X+Y \leq 1) = f(0,0) + f(0,1) + f(1,0)$$

$$= 3/28 + 6/28 + 9/28$$

$$P(X+Y \leq 1) = 18/28$$

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3.49

Let X denote the number of times a certain numerical control machine will malfunction 1, 2 or 3 times on any given day. Let Y denote the number of times a Technician is called on an emergency call. Their joint ~~func~~ probability distribution is given:

		X		
		1	2	3
Y	1	0.05	0.05	0.1
	2	?	0.1	0.35
	3	0	0.2	0.1

i) Evaluate the marginal distrib of X
 ii) Evaluate the marginal of Y
 iii) $P(Y=3/X=2)$

Soln:

X	$g(x)$
1	0.1
2	0.35
3	0.55

$\sum g(x) = 1$
 P.d.f

Y	$h(y)$
1	0.2
2	0.50
3	0.3

$\sum h(y) = 1$
 P.d.f

$P(Y=3/X=2) = \frac{P(X \cap Y)}{P(X)} = \frac{0.2}{0.35} = 0.57$

Joint Probability Distribution

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8.51 Consider an Experiment that Consists of 2 rolls of a balanced die. If X is the number of 4's and Y is the number of 5's obtained in the 2 rolls of the die. find

a) the joint probability Distribution.

6	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)
5	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)
4	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)
3	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)
2	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
1	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)
	1	2	3	4	5	6

$0 \leq x+y \leq 2$

$x = \text{number of 4's} = 0, 1, 2$

$y = \text{number of 5's} = 0, 1, 2$

$x \backslash y$	0	1	2	$P(x,y)$
0	(0,0) $1/36$	(0,1) $2/36$	(0,2) $1/36$	$4/36$
1	(1,0) $2/36$	(1,1) $4/36$	(1,2) $2/36$	$8/36$
2	(2,0) $1/36$	(2,1) $2/36$	(2,2) $1/36$	$4/36$
$P(x)$	$4/36$	$8/36$	$4/36$	1

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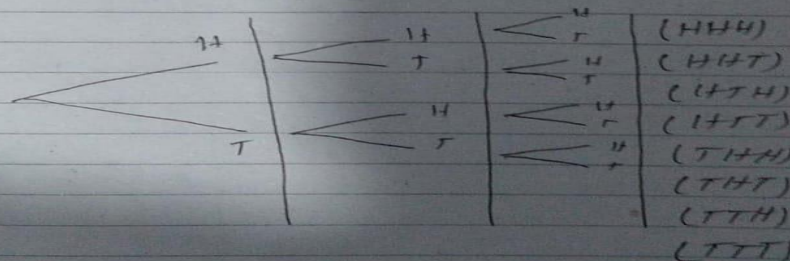
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$$\begin{aligned} b) \quad P\{(X,Y) \in A\} &= P\{2X+Y < 7\} \\ &= f(0,0) + f(0,1) + f(0,2) + f(1,1) \\ &= \frac{16}{36} + \frac{4}{36} + \frac{1}{36} + \frac{8}{36} \\ &= \frac{29}{36} \end{aligned}$$

3.52. let X denote the number of heads and Y denote the number of ~~tails~~ heads minus the number of tails. When a fair coin is tossed 3 times. Find the Joint probability Distribution of X and Y .

Solution

Sample Space.



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S.S.	x	$y = H - T$	
HHH	3	3	$1/8$
HHT	2	1	$3/8$
HTH	2	1	$3/8$
HTT	1	-1	$1/8$
THH	2	+1	
THT	1	-1	
TTH	1	-1	
TTT	0	-3	

$y \backslash x$	0	1	2	3
-3	(0, -3)	(1, -3)	(2, -3)	(3, -3)
-1	(0, -1)	(1, 1)	(2, -1)	(3, -1)
1	(0, 1)	(1, 3)	(2, 1)	(3, 1)
3	(0, 3)	(1, 5)	(2, 3)	(3, 5)
	0	1	2	3
-3	(0, -3)	(1, -3)	(2, -3)	(3, -3)
	$1/8$			
-1	(0, -1)	(1, 1)	(2, -1)	(3, -1)
		$3/8$		
1	(0, 1)	(1, 3)	(2, 1)	(3, 1)
			$3/8$	
3	(0, 3)	(1, 5)	(2, 3)	(3, 5)
				$1/8$

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3.53 Three Cards are drawn with out Replacement from 12 face Cards (J, K, Q) of ordinary deck of 52 playing Cards.
Let X is o.v for number of Kings.
 Y is o.v for number of Jacks.

a) Find Joint Probability Distribution.
b) $P\{(X, Y) \in A\} \Rightarrow A = \{(x, y) / x + y \geq 2\}$

$X = \text{o.v} = \text{King}$
 $Y = \text{o.v} = \text{Jack}$

$0 \leq x + y \leq 3$

$X = 0, 1, 2, 3$
 $Y = 0, 1, 2, 3$

$$f(x, y) = \frac{\binom{4}{x} \binom{4}{y} \binom{4}{3-x-y}}{\binom{12}{3}}$$

$f(x, y) =$

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		x				
		0	1	2	3	$h(y)$
y	0	(0,0) $1/55$	(1,0) $6/55$	(2,0) $6/55$	(3,0) $1/55$	$14/55$
	1	(0,1) $6/55$	(1,1) $16/55$	(2,1) $6/55$	(3,1) $1/55$	$28/55$
	2	(0,2) $6/55$	(1,2) $6/55$	(2,2) $1/55$	(3,2) $1/55$	$13/55$
	3	(0,3) $1/55$	(1,3) $1/55$	(2,3) $1/55$	(3,3) $1/55$	$4/55$
$g(x)$		$14/55$	$28/55$	$13/55$	$4/55$	1

b) $P[(X,Y) \in A] \quad A \Rightarrow \{(x,y) \mid x+y \geq 2\}$

$$\Rightarrow 1 - P(x+y < 2)$$

$$\Rightarrow 1 - (f(0,0) + f(1,0) + f(0,1))$$

$$\Rightarrow 1 - (1/55 + 6/55 + 6/55)$$

$$\Rightarrow 1 - 13/55 = \frac{55-13}{55}$$

$$\Rightarrow \frac{42}{55}$$